

SMART CITY BIKE SHARING ANALYTICS USING EXCEL & POWER BI

1. Project Overview & Objective

Project Overview

This project focuses on analysing a Smart City Bike Sharing dataset using Microsoft Excel and Power BI. The dataset was cleaned, transformed, and prepared in Excel before being imported into Power BI for data modelling, DAX calculations, and interactive dashboard creation. The dashboard provides valuable insights into bike station availability, bike stands, station status, banking facilities, and bonus-enabled stations.

Objective

- ✓ Clean and transform raw bike sharing data using Excel.
- ✓ Perform data modelling and DAX calculations in Power BI.
- ✓ Develop an interactive dashboard for monitoring bike station performance.
- ✓ Analyse station availability, infrastructure, and operational status to support better business decisions.

2. Data Source

 **Source:** Open Government Smart City Bike Sharing Dataset

 **Domain:** Transportation / Smart City / Public Mobility Analytics

 **Dataset Type:** Bike Sharing Station Information

Dataset Includes

-  Bike Stations
-  Available Bikes
-  Bike Stands
-  Banking Facility
-  Bonus Facility
-  Station Status
-  Geographic Position
-  Last Update Date & Time

? 3. Problem Statement

This project aims to analyse bike sharing station performance and operational efficiency by answering the following business questions:

- ✓ 🚲 How many bike stations are available?
- ✓ 🚲 Which cities have the highest number of available bikes?
- ✓ 🅑 Which cities have the highest bike stand capacity?
- ✓ 🇮🇹 What is the overall bike availability percentage?
- ✓ 🏦 How many stations support Banking facilities?
- ✓ 🎁 How many stations support Bonus facilities?
- ✓ 🟢 What is the operational status (Open/Closed) of stations?
- ✓ 📍 Where are the stations geographically located?

📄 4. Attribute Details

🚲 Attribute	📁 Data Type	📖 Description
🚲 Name	Text	Bike Station Name
🔢 Number	Integer	Station ID
📍 Address	Text	Station Address
🌐 Contract Name	Text	City/Region
🏦 Banking	Boolean	Banking Facility
🎁 Bonus	Boolean	Bonus Facility
🅑 Bike Stands	Integer	Total Bike Stand Capacity
🚲 Available Bikes	Integer	Bikes Available
🚲 Available Bike Stands	Integer	Empty Bike Stands
🟢 Status	Text	Open / Closed

 Attribute	 Data Type	 Description
 Position	Geographic	Latitude & Longitude
 Last Update Date	Date	Last Updated Date
 Last Update Time	Time	Last Updated Time

5. Tools & Technologies

Microsoft Excel

- ✓  Data Cleaning
- ✓  Data Transformation
- ✓  Pivot Tables
- ✓  Data Validation

Microsoft Power BI

- ✓  Data Modelling
- ✓  DAX Measures
- ✓  Dashboard Development
- ✓  Maps & KPIs
- ✓  Interactive Slicers

6. Data pre-processing

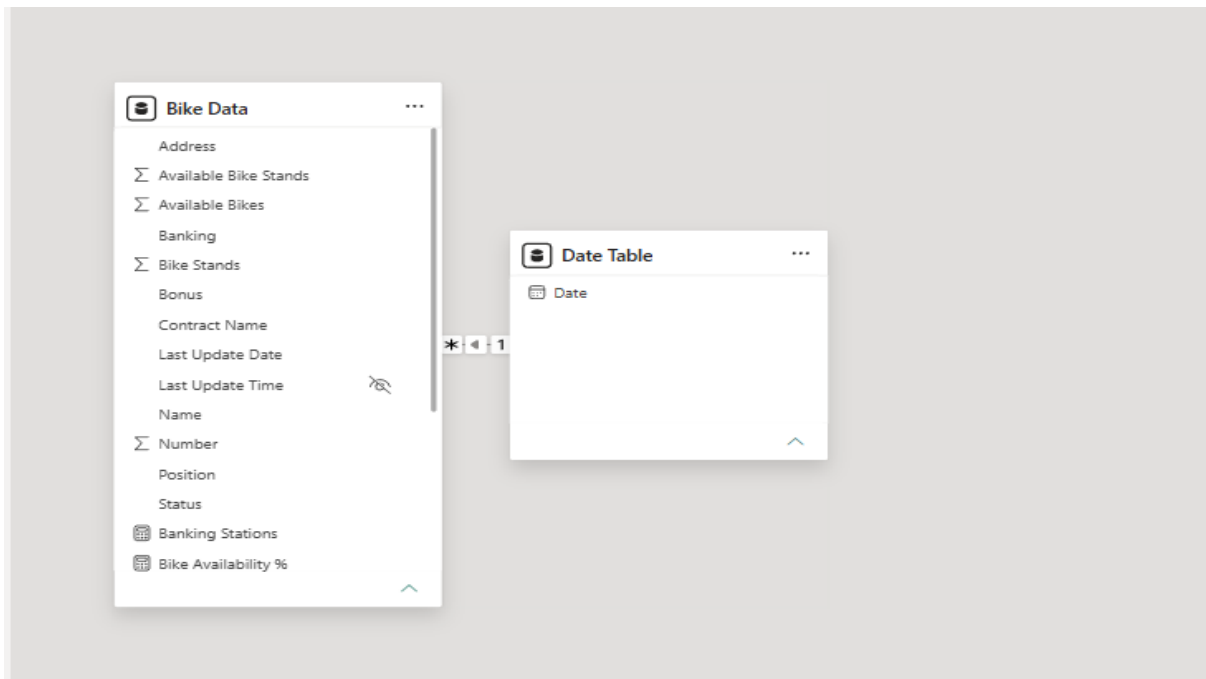
Tasks Performed

- ✓  Removed unwanted characters
- ✓  Checked duplicate records
- ✓  Handled missing values
- ✓  Converted Banking & Bonus into Boolean values
- ✓  Split Date & Time columns
- ✓  Standardised column names
- ✓  Verified data types
- ✓  Created Date Table

🔗 7. Data Modelling & DAX

⚙️ Data Model

- ✓ Created Date Table
- ✓ Established relationship between tables
- ✓ Implemented Star Schema (where applicable)



📊 DAX Measures

- ✓ 🚲 Total Stations

```
1 Total Stations =  
2 COUNT('Bike Data'[Number])
```

- ✓ 🚲 Total Available Bikes

```
1 Total Available Bikes =  
2 SUM('Bike Data'[Available Bikes])
```

- ✓ 📊 Total Bike Stands

```
1 Total Available Bike Stands =  
2 SUM('Bike Data'[Available Bike Stands])
```

✓ 📊 Bike Availability %

```
1 Bike Availability % =  
2 DIVIDE(  
3 'Bike Data'[Total Available Bikes],  
4 'Bike Data'[Total Bike Stands],  
5 0  
6 )
```

✓ ● Open Stations

```
1 Open Stations =  
2 CALCULATE(  
3 COUNT('Bike Data'[Number]),  
4 'Bike Data'[Status]="OPEN"  
5 )
```

✓ ● Closed Stations

```
1 Closed Stations =  
2 CALCULATE(  
3 COUNT('Bike Data'[Number]),  
4 'Bike Data'[Status]="Closed"  
5 )
```

✓ 🏠 Banking Stations

```
1 Banking Stations =  
2 CALCULATE(  
3 COUNT('Bike Data'[Number]),  
4 'Bike Data'[Banking]=TRUE()  
5 )
```

✓ 🎁 Bonus Stations

```
1 Bonus Stations =  
2 CALCULATE(  
3 COUNT('Bike Data'[Number]),  
4 'Bike Data'[Bonus]=TRUE()  
5 )
```

📊 8. Analysis & Visualizations

📈 Dashboard Components

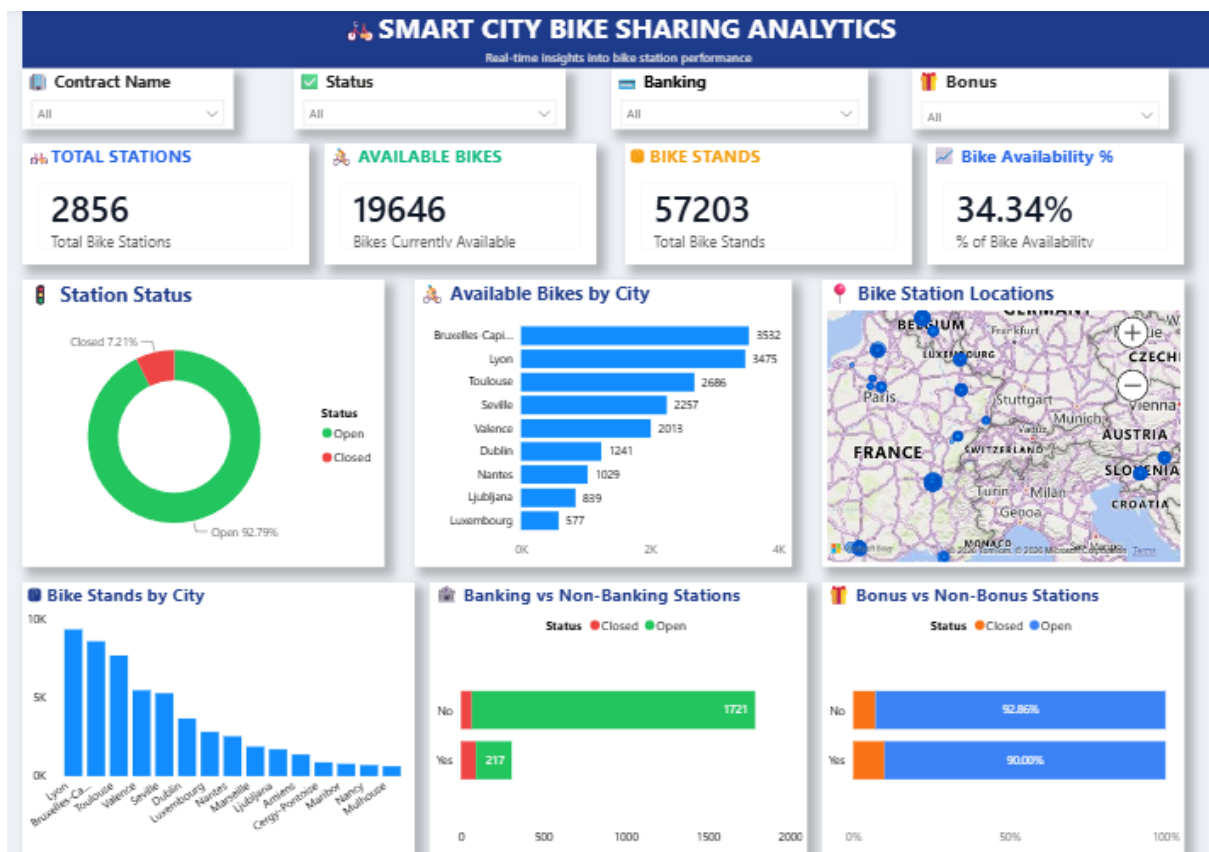
📌 KPI Cards

- 🚲 Total Stations
- 🚲 Available Bikes
- 🅅 Bike Stands
- 📊 Bike Availability %

- ✦ 📊 Station Status (Donut Chart)
- ✦ 📊 Available Bikes by City (Bar Chart)
- ✦ 📊 Bike Stands by City (Column Chart)
- ✦ 📍 Bike Station Locations (Map)
- ✦ 🏠 Banking vs Non-Banking Stations
- ✦ 🎁 Bonus vs Non-Bonus Stations

🛠️ Interactive Features

- ✓ 📊 Slicers
 - 🌐 Contract Name
 - 🟢 Status
 - 🏠 Banking
 - 🎁 Bonus
- ✓ 🔄 Cross Filtering
- ✓ 📄 Interactive Dashboard



💡 9. Insights & Conclusions

📌 Key Metrics

📍 Total Bike Stations: **2,856**

🚲 Total Available Bikes: **19,646**

🅑 Total Bike Stand Capacity: **57,203**

📊 Overall Bike Availability: **34.34%**

🔍 Key Observations

- ✓ Most stations are currently operational.
- ✓ Bike availability differs significantly across cities.
- ✓ Banking and Bonus facilities are available only at selected stations.

🔧 Improvement Opportunities

- 📊 High-demand cities may require additional bike allocation.
- 📊 Stations with consistently low availability should be monitored closely.

🎯 Recommendations

- ✓ 🚲 Increase bikes in high-demand areas.
- ✓ 📍 Improve monitoring of low-availability stations.
- ✓ 🏠 Expand Banking-enabled stations.
- ✓ 🎁 Increase Bonus-enabled stations to improve customer experience.

✅ 10. Conclusion

The Smart City Bike Sharing Analytics project demonstrates a complete end-to-end analytics workflow using Microsoft Excel and Power BI. The project successfully transformed raw data into an interactive dashboard, providing actionable insights into station performance, bike availability, infrastructure, and operational efficiency. The dashboard enables stakeholders to make informed, data-driven decisions for effective bike sharing management.

